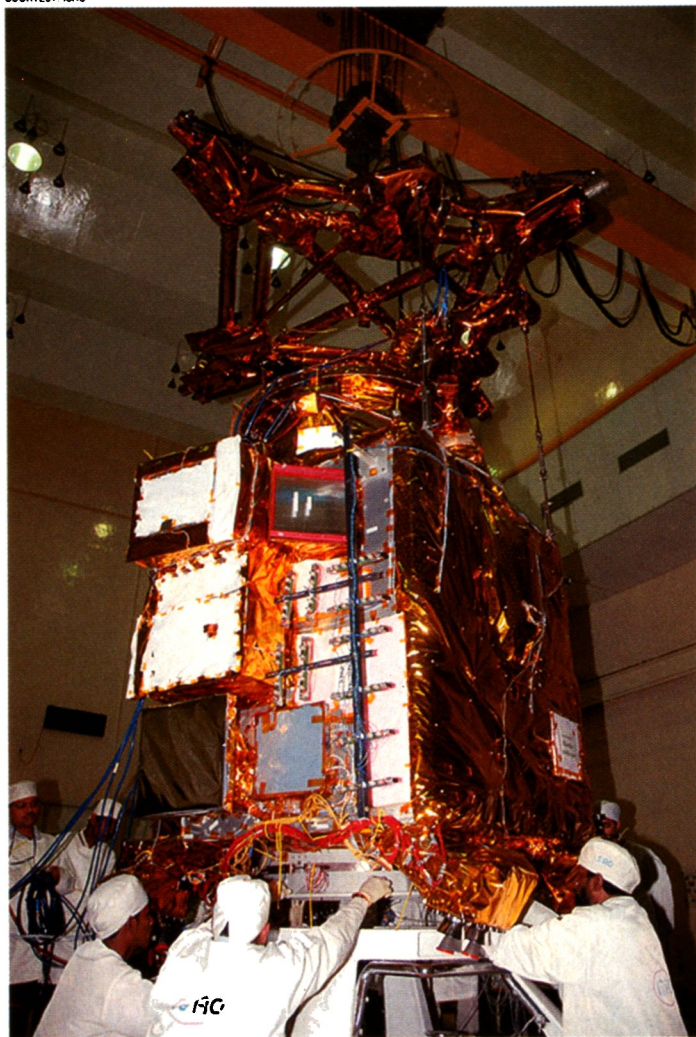


COURTESY: ISRO



MOTHERLOAD:
Chandrayaan
undergoing
tests in
Bangalore be-
fore its launch

IF ALL GOES WELL, India's lunar mission would provide the world with the best digital global map of the moon than it has ever had.

which would map the moon's surface for water ice in its dark polar regions. Bulgaria has a Radiation Dose Monitor (RADOM) on board to measure the radiation levels on the lunar surface. The Swedish and the Japanese have collaborated to put together an analyser that will measure the effects of solar wind on the moon. While the UK has put an X-ray spectrometer that would map its mineral resources, Germany's Max Planck Institute is flying an infra-red spectrometer to study lunar mineralogy.

In all, Chandrayaan would have 11 scientific payloads on board, making it the most instrumented lunar orbiter ever. It also has to survive the extreme temperatures of heat and cold that it would experience as it orbits the moon for the next two years. Paul Spudis, principal investigator for NASA's Miniature Synthetic Aperture Radar or MiniSAR that is on board the Chandrayaan, says: "If everything works, Chandrayaan would provide the world with the best digital global map of the moon than it has ever had. Even if half the instruments on board function, it would still be a tremendous success."

What Spudis is saying is surprising but true. Despite over 67 lunar missions being launched ever since the Soviets began the race with a fly-by in 1959, the moon remains a poorly studied planetary body. A deadly race broke out between America and the Soviet Union—they first competed by putting out robotic probes on the moon. When the US found the Soviet streets ahead of them in the early '60s, President John F. Kennedy declared that America would put a man on the moon by the end of that decade. The

have an impact probe to mark India's presence on the moon. As he told Nair, "If you are going to go that far, why not at least have an impactor."

Then over the next few days, Chandrayaan's instruments would start functioning. Apart from the moon impact probe, ISRO has four other scientific instruments that include terrain-mapping cameras, ultra-sophisticated imagers and spectrometers that would send back tonnes of data about the moon's terrain, topography, mineral resources and chemical composition.

Chandrayaan is also carrying six payloads from abroad. NASA is sending two probes, one of

US achieved that when Neil Armstrong and Buzz Aldrin became the first humans to walk on the moon on July 20, 1969.

After the moon fever wore off during mid '70s, America ended all its missions and went back with an orbiter only in 1994. Soviet Union too dropped its moon missions by 1976 never to return. But the '90s saw a major revival of interest in the moon with reports of water ice being found and also abundant quantities of Helium-3, which was being regarded as the fuel of the future. After Japan sent its Hiten orbiter in 1990, the US and the European Space Agency followed